Module 3
Universal health coverage in the context of aging

1. Overview and concepts

Overview

1. Overview and concepts
2. Financial risk protection I
3. Financial risk protection II
4. Health Service Coverage
5. Equity
6. Aging
7. Example 1: Japan
8. Example 2: Bangladesh
9. Future policy directions

Definition

Everyone who needs health services is able to access them, without undue financial hardship.

Source: Boerma T, Eozenou P, Evans D, Evans T, Kieny M-P, Wagstaff A (2014) Monitoring Progress towards Universal Health Coverage at Country and Global Levels. PLoS Med 11(9): e1001731.

Concepts I

Access to health services

- Availability
- Quality
- Proximity

Effective coverage of interventions

- Timeliness
- Quality

Source: Boerma *et al.* 2014

Concepts II

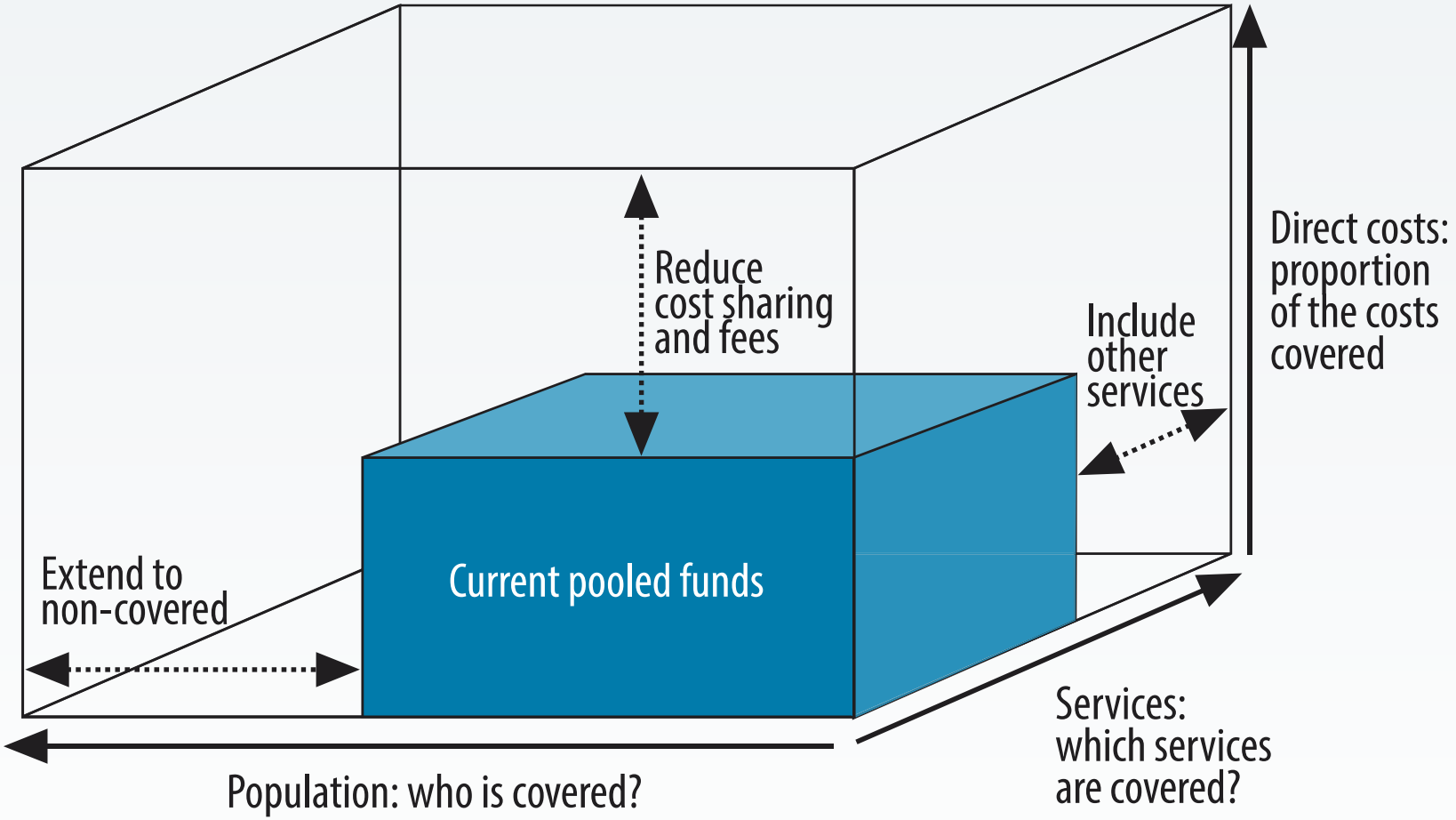
Financial risk protection

- Protection against financial hardship

Equity

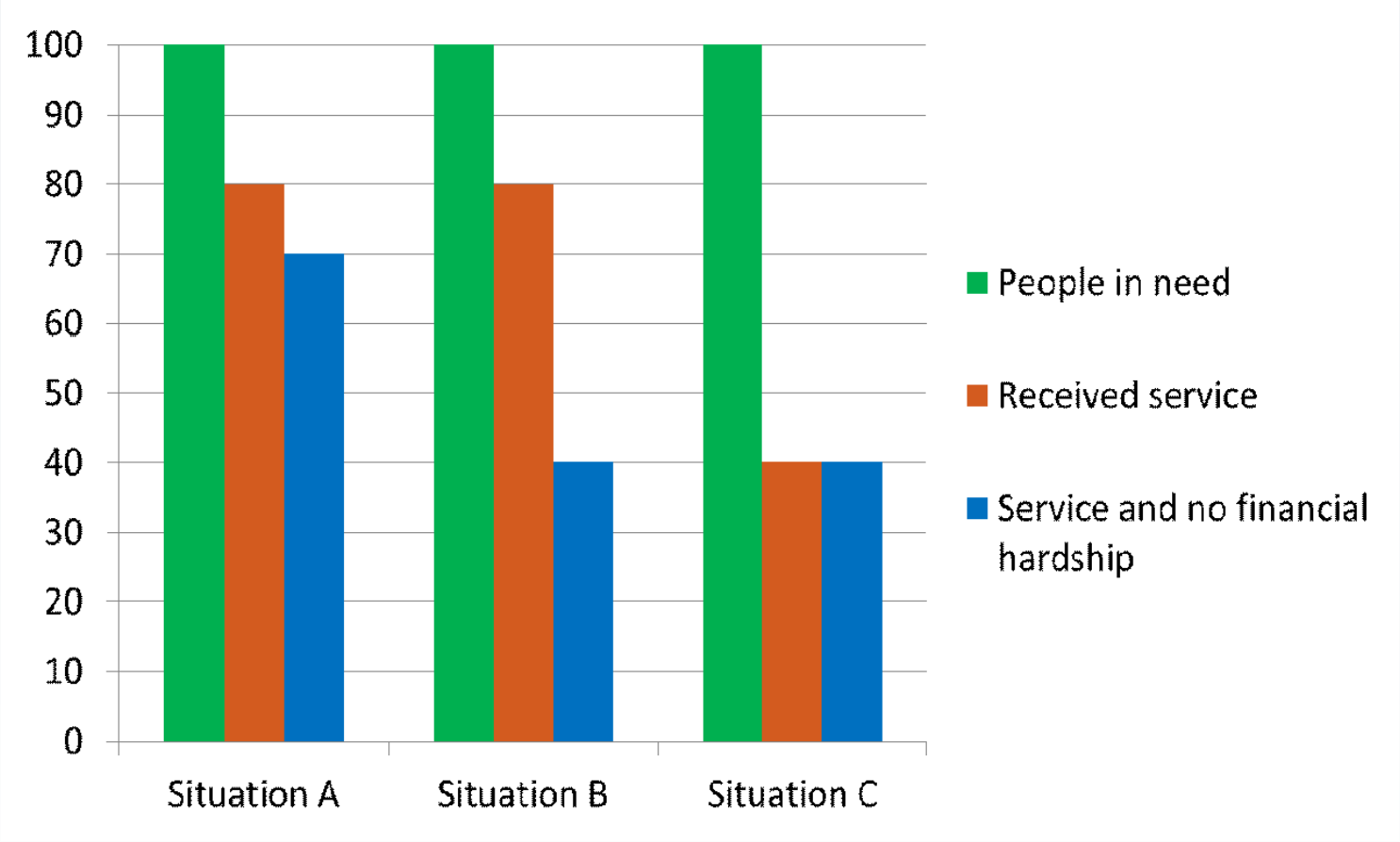
- People get what they need at an affordable price

Dimensions

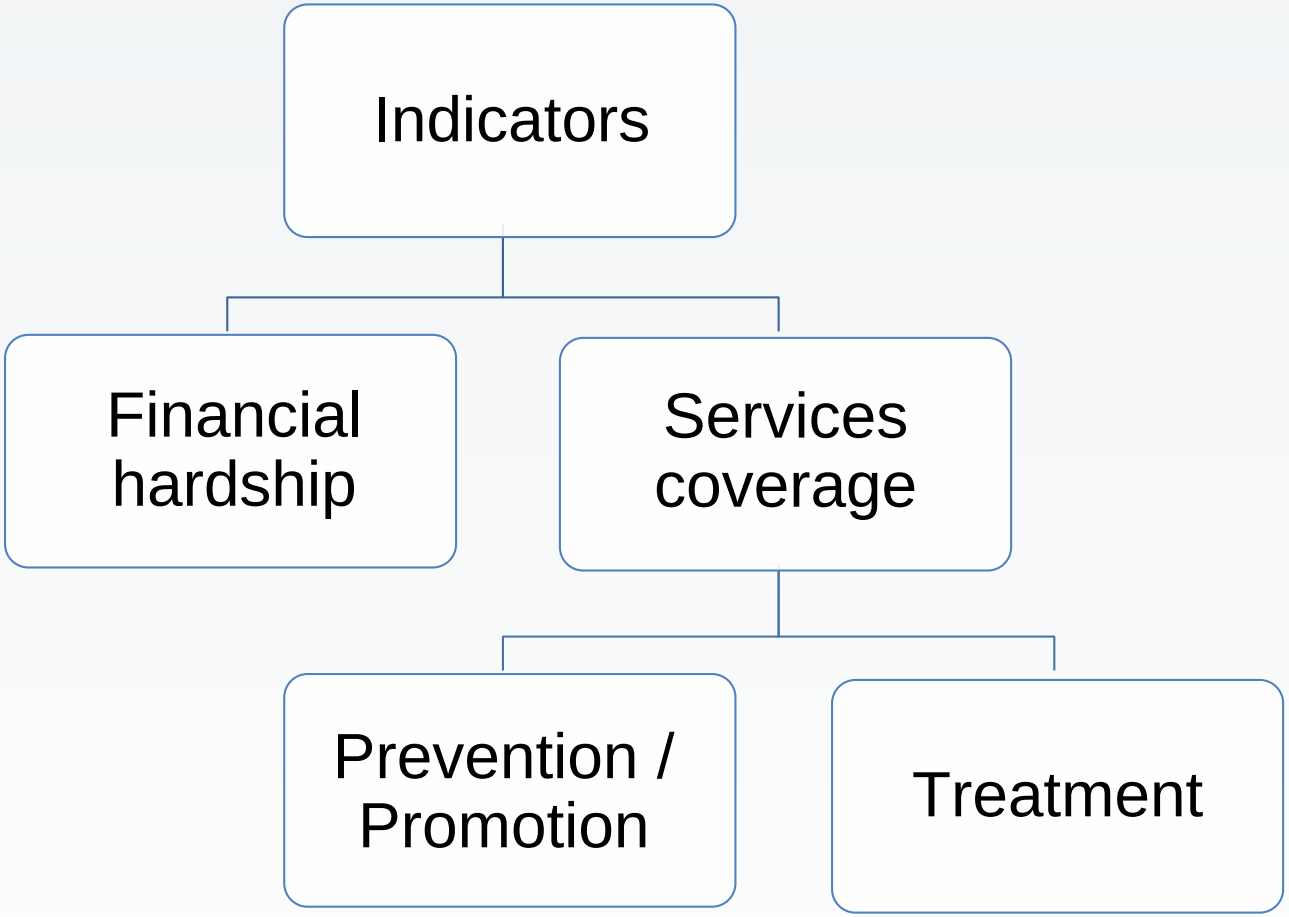


Source: World Health Organization (2010) The world health report: health systems financing: the path to universal coverage.

Hypothetical examples of combined summary measure results



Tracking areas



Summary

Concepts

- Access, coverage, risk protection and equity

Dimensions

- Population, services and direct costs

Tracking areas

- indicators, financial hardship and service coverage

2. Financial Risk Protection I

Indicators of financial protection I

Incidence of catastrophic payment

- Proportion of households facing excessive financial burden due to out-of-pocket (OOP) health payments

Mean positive catastrophic overshoot

- Extent to which household spending on health exceeds the threshold for catastrophic health expenditure

Indicators of financial protection II

Incidence of impoverishment

- Proportion of households driven below the poverty line due to OOP payment

Intensity of poverty

- Amount by which a household fell further into poverty due to health spending

Catastrophic payment: measurement

$$\frac{\text{OOP payment}}{\text{Household wealth}} = \text{Catastrophe criterion}$$

Thresholds: capacity to pay (≥40%), non-food (>25 or 40%) and total consumption (>10 or 15%)

Example: Catastrophic payments

	Assume share spent on health	Incidence of catastrophic payment	Overshoot
Household 1	45%	1	35%
Household 2	30%	1	20%
Household 3	20%	1	10%
Household 4	10%	0	0%
Household 5	5%	0	0%
Total (%)		3/5=60%	65%
Mean overshoot(%)			65/5=13%
Mean <u>positive</u> overshoot (%)			65/3=21.7%

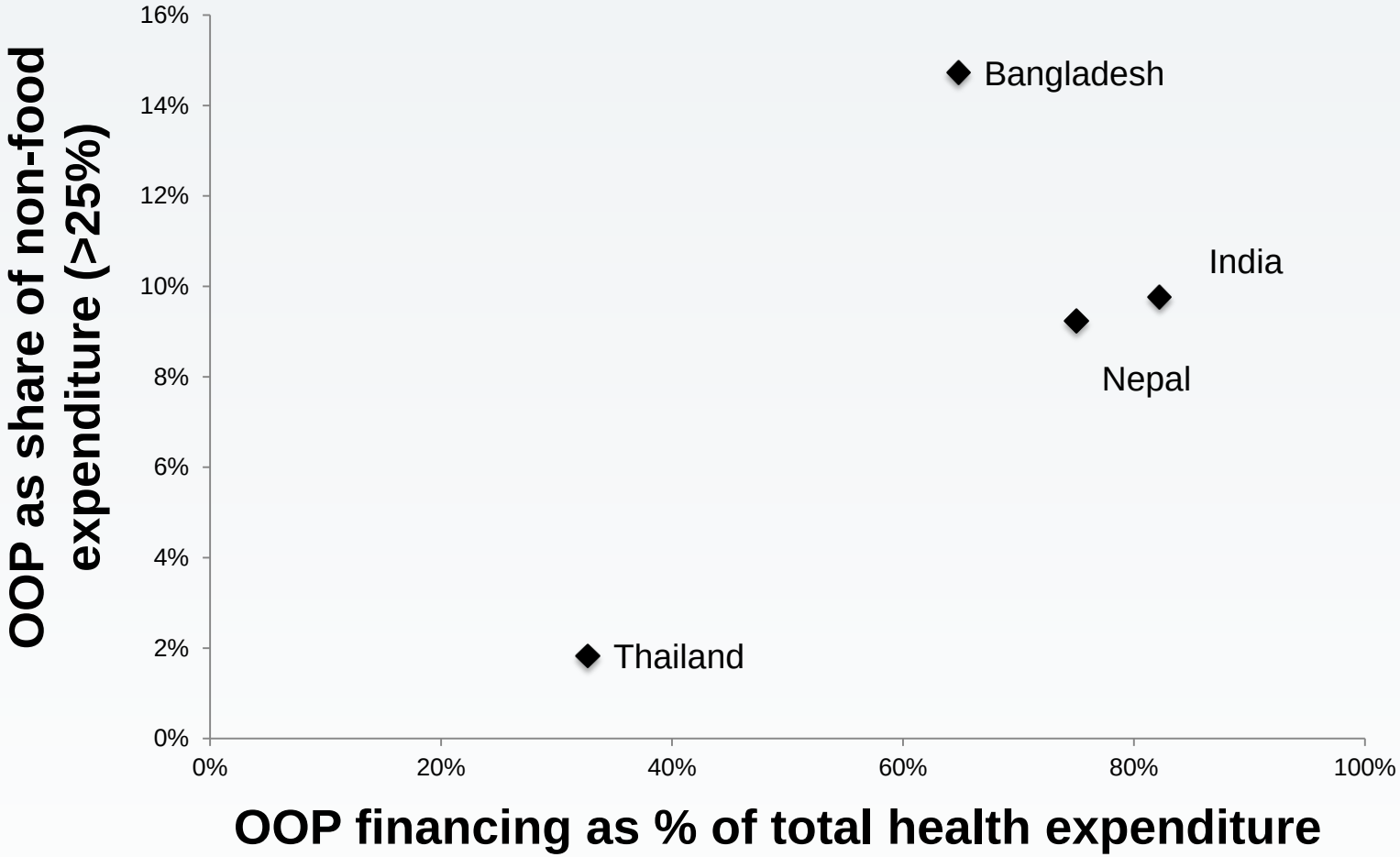
Assumes catastrophic payment defined at threshold of 10% of household consumption

Example: Bangladesh

Definitions and thresholds	Percent	95% Confidence Interval
Total consumption		
>15%	14.9	12.7 – 17.4
Non-food		
>25%	17.6	14.8 – 20.7
>40%	9.9	8.0 – 2.1
Capacity to pay		
≥40%	9.0	7.2 – 11.2

Source: Rahman MM, Gilmour S, Saito E, Sultana P, Shibuya K (2013) Health-Related Financial Catastrophe, Inequality and Chronic Illness in Bangladesh. PLoS ONE 8(2): e56873. CC-BY-4.0.

Example: Developing countries



Source: van Doorslaer *et al*. (2007), Catastrophic payments for health care in Asia. Health Econ., 16: 1159–1184.

Summary

Indicators of financial protection

- Incidence of catastrophic payment
- Mean positive catastrophic overshoot
- Incidence of impoverishment
- Intensity of poverty

Catastrophic payment

- OOP payment/household wealth=criterion

3. Financial Risk Protection II

Questions

Who is covered?

- Population enrolled (% of total)

How much is covered? (Scope of services)

- Comprehensive
- Inpatient

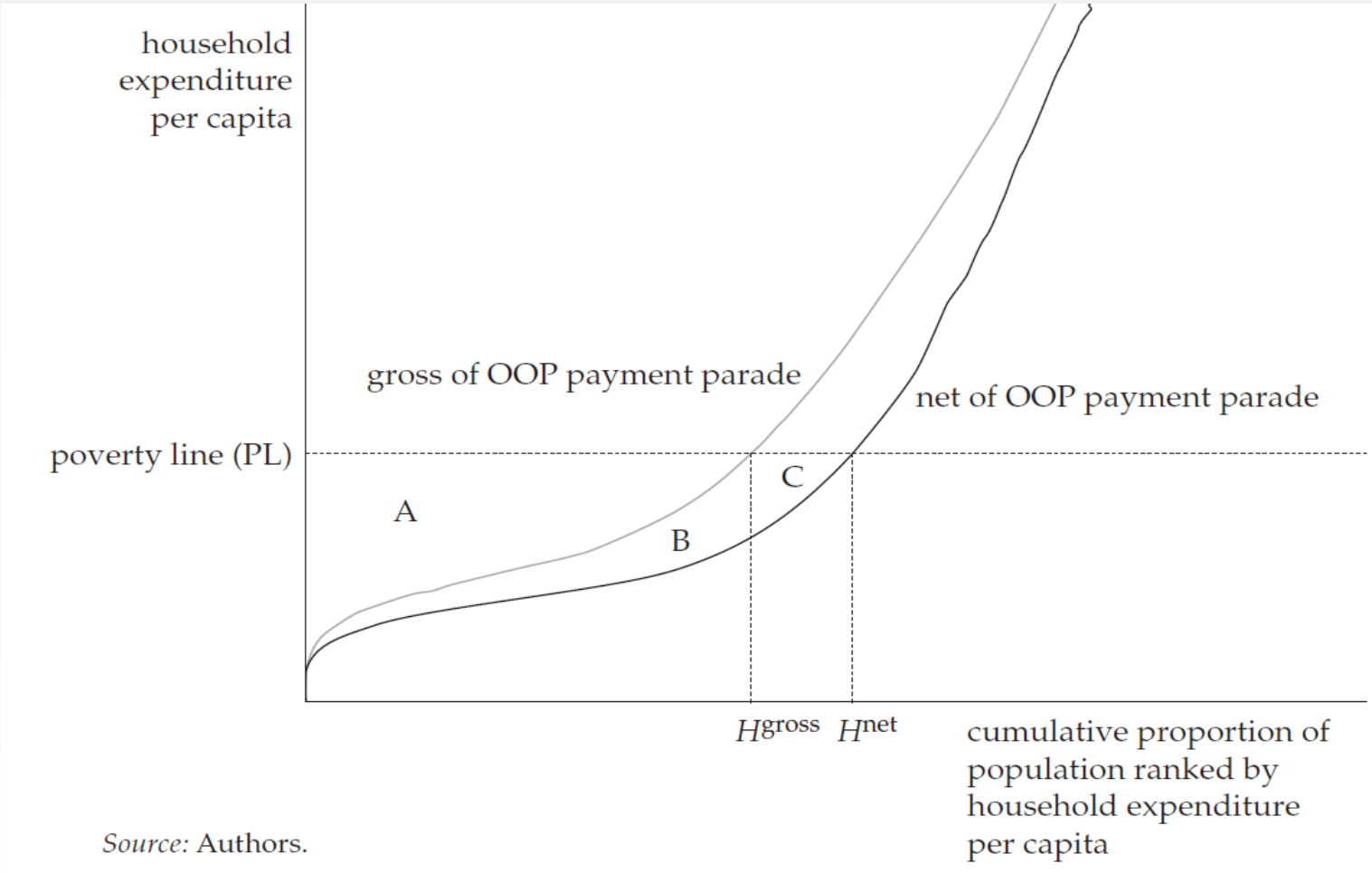
What % total health expenditure is covered by OOP payments?

Example: Insurance and OOP

Who is covered?		What is covered?	How much is covered?		% Change in OOP since reform
	Population(s) targeted by health insurance	Population enrolled (% of total)	Scope of services	OOP as % of total health exp. in 2010	
Intermediate-stage reform countries					
Ghana	Entire population	54	Comprehensive	27	-4
Philippines	Entire population	76	Inpatient, with outpatient for poor people	54	4
Rwanda	Entire population	92	Comprehensive	22	-3
Vietnam	Entire population	42	Comprehensive	58	-6
Early-stage reform countries					
Mali	Entire population	3	Comprehensive	53	1

Source: Lagomarsino, G., Garabrant, A., Adyas, A., Muga, R., & Otoo, N. (2012). Moving towards universal health coverage: health insurance reforms in nine developing countries in Africa and Asia. The Lancet, 380(9845), 933-943.

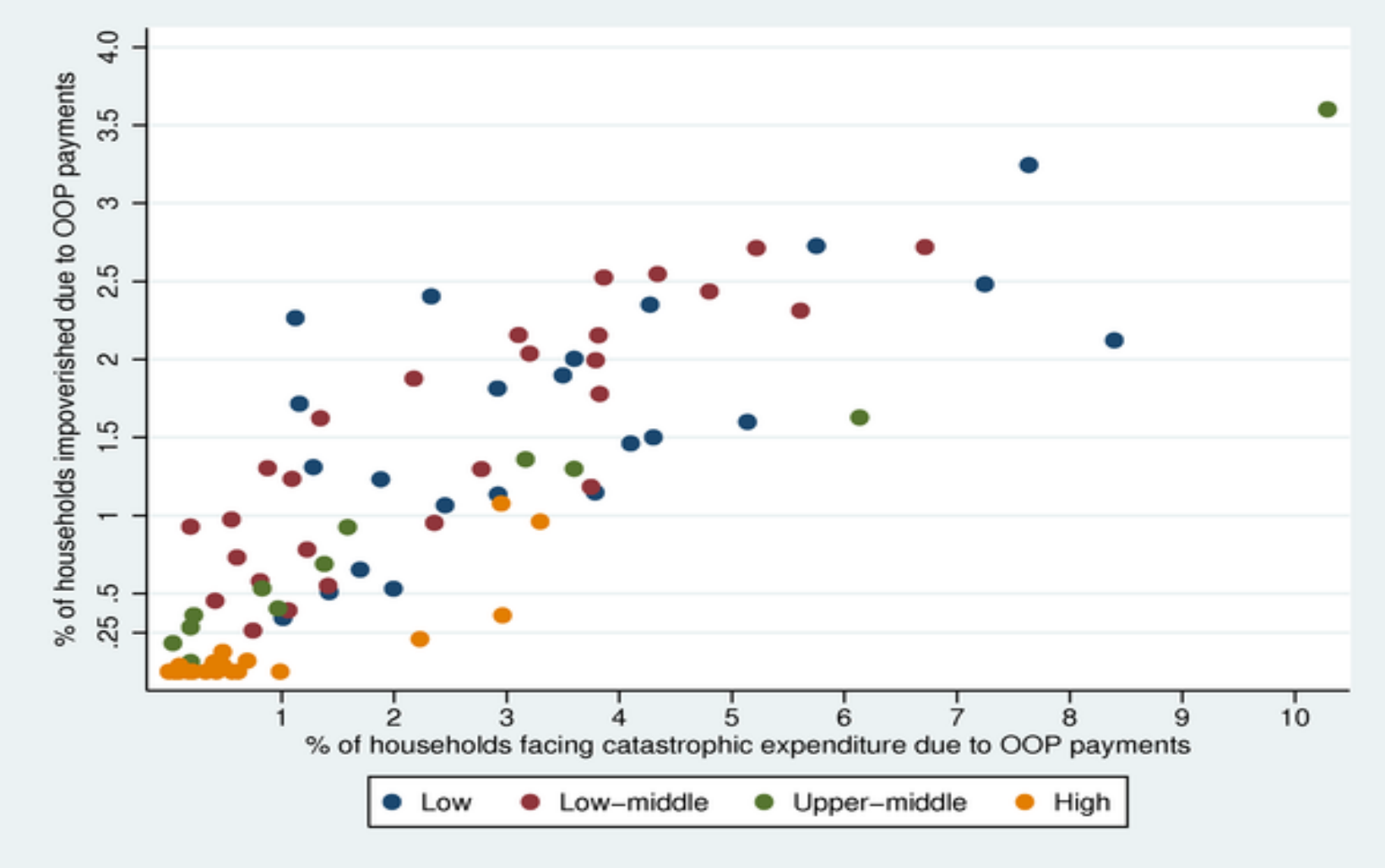
Impoverishment: Measurement



Pen's Parade for Household Expenditure Gross and Net of OOP Health Payments

Source: O'Donnell, O. A., & Wagstaff, A. (2008). Analyzing health equity using household survey data: a guide to techniques and their implementation. World Bank Publications. CC BY-3.0 IGO.

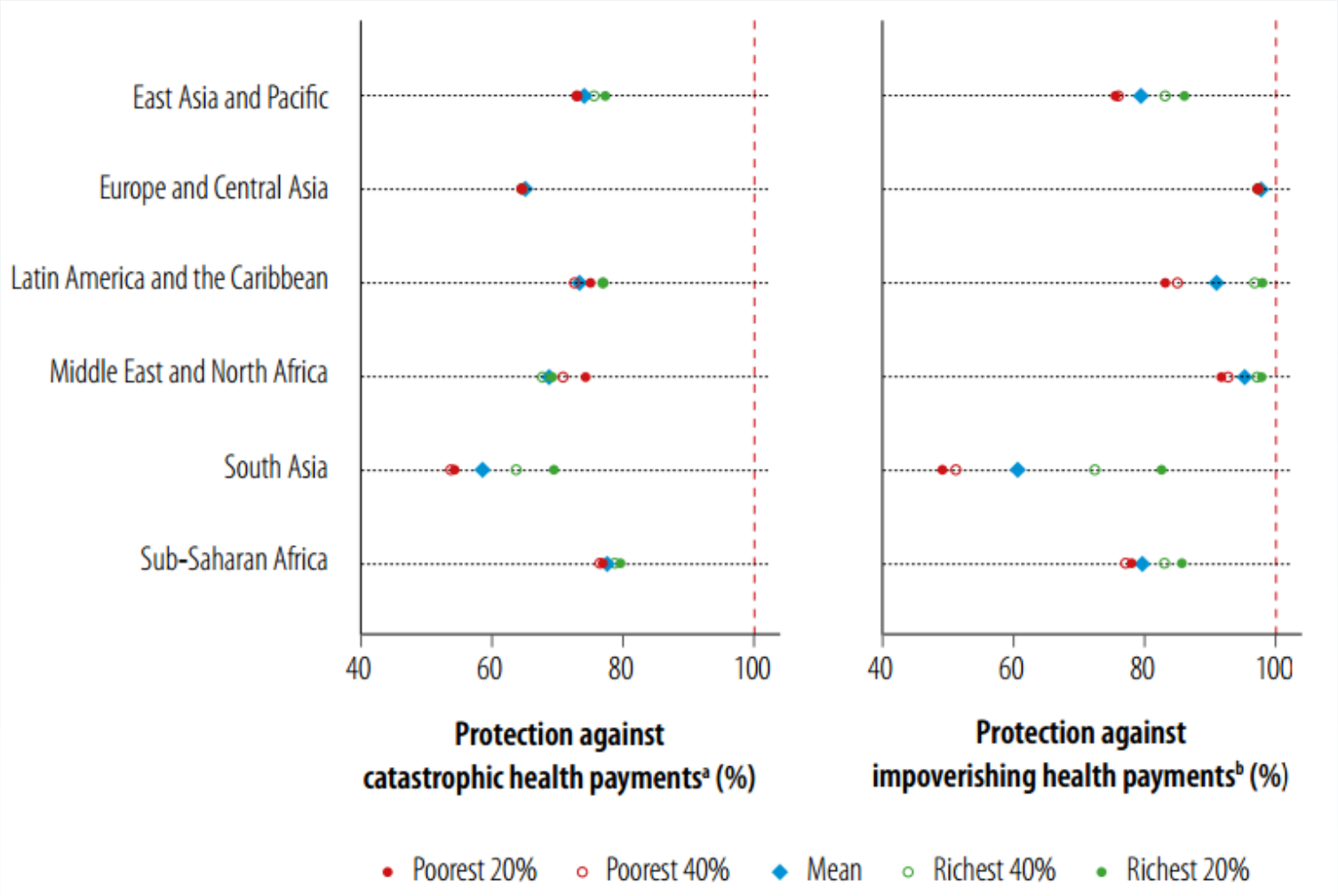
Impoverishment and catastrophic health expenditure



Impoverishment and catastrophic health expenditure headcount by country income

Source: Saksena P, Hsu J, Evans DB (2014) Financial Risk Protection and Universal Health Coverage: Evidence and Measurement Challenges. PLoS Med 11(9): e1001701. doi:10.1371/journal.pmed.1001701

Financial risk protection by region



Financial protection measures against catastrophic and impoverishing health payments, by region

Source: World Health Organization and The World Bank (2014) Monitoring Progress towards Universal Health Coverage at Country and Global Levels: Framework, Measures and Targets. Joint WHO/World Bank Group paper.

Summary

Financial risk assessment

- Population, services and cost

Impoverishment measurement

Global variation

- Most severe in Southeast Asia
- Biggest problem is impoverishment

4. Health Service Coverage

Criteria for indicator selection

Relevance

- Addresses health priority
- Cost-effective intervention

Quality

- "Effective" coverage

Availability

- Standard methods
- Measured on regular basis
- Comparable

Equity

- Disaggregation possible

Coverage and effective coverage

Coverage

- Share of a population receiving a set of interventions

Effective coverage

- Services that achieve those health outcomes

Example for hypertension

- Ratio of actual systolic blood pressure reduction from treatment of hypertension to potential blood pressure reduction

Grouping of indicators

Promotion / Prevention	Treatment
Improved water	ARI and diarrhea
Adequate sanitation	Skilled birth attendants
DPT3, MSL, BCG, Polio	Institutional delivery
ANC4+ from SBP	Diabetes control
Seeking care for ARI and diarrhea	Hypertension control
No use of solid fuels	
No use of tobacco	

Measurement of services coverage

Promotion/Prevention example

- Proportion of households with improved water

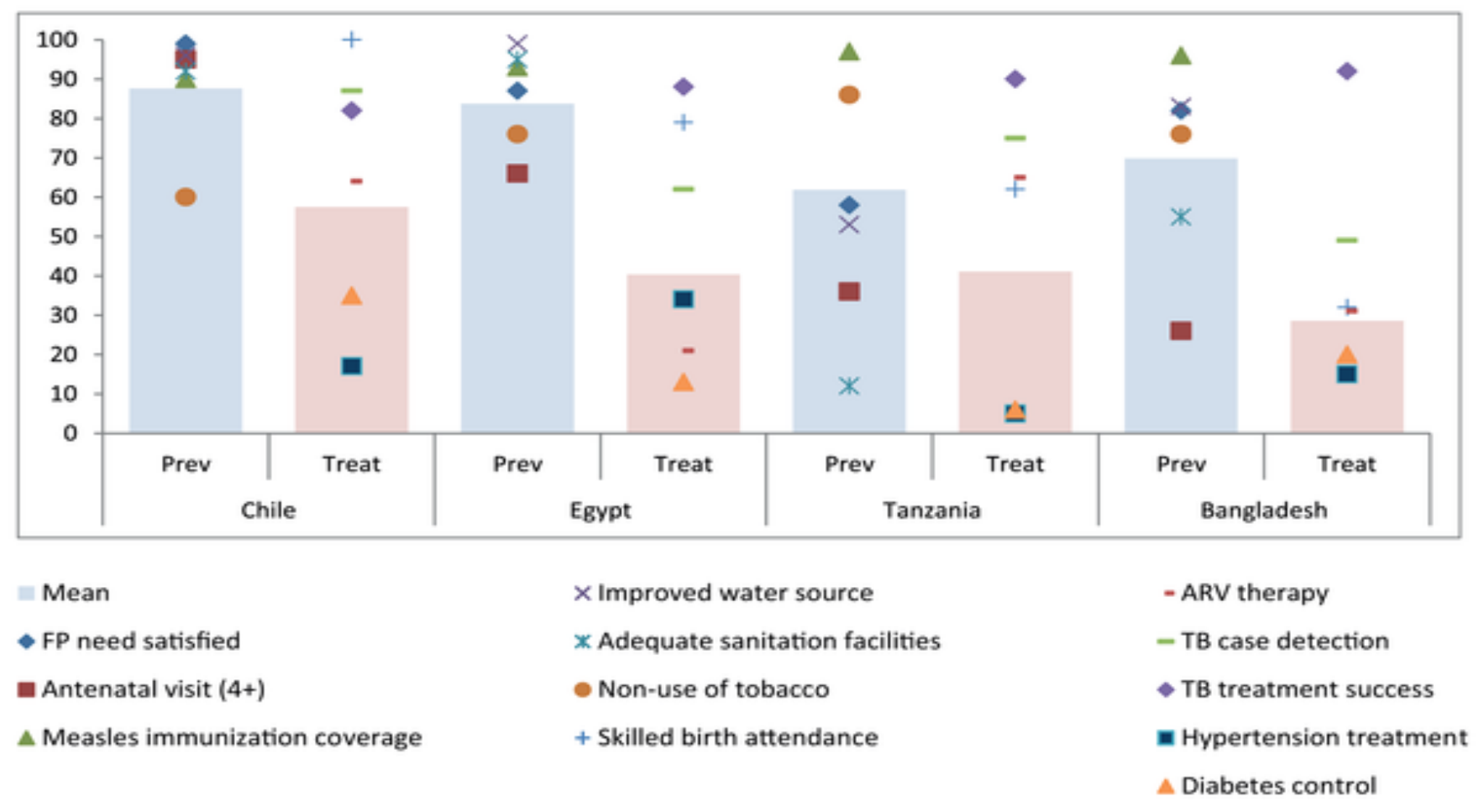
Treatment example

- Proportion of deliveries assisted by a skilled birth attendant

Measurement

- Weighted mean: regression
- Unweighted mean: simple mean

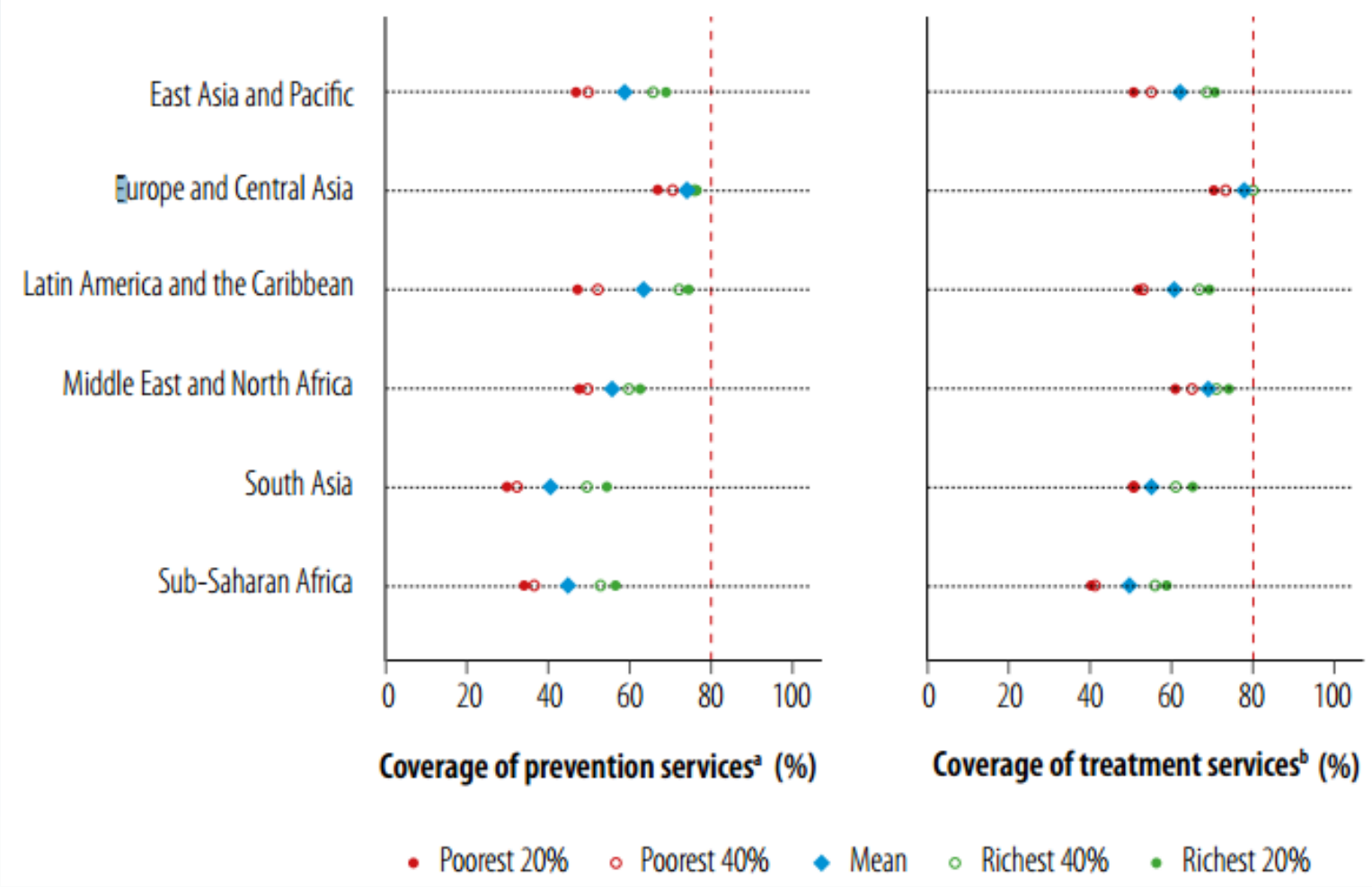
Prevention and treatment coverage



Service coverage rates for six illustrative prevention interventions and six illustrative treatment interventions for four countries

Source: The World Health Organization and The World Bank (2014) Monitoring Progress towards Universal Health Coverage at Country and Global Levels: Framework, Measures and Targets. Joint WHO/World Bank Group paper.

Service coverage by region



Coverage of prevention and treatment services, by region

Source: The World Health Organization and The World Bank (2014) Monitoring Progress towards Universal Health Coverage at Country and Global Levels: Framework, Measures and Targets. Joint WHO/World Bank Group paper.

Summary

Criteria for selecting indicators

- Relevance, quality, access and equity

Intervention categories

- Prevention coverage higher than treatment

Global service coverage

- Variation in coverage gap between rich and poor

5. Equity

Definition

Everyone who needs health services is able to access them, without undue financial hardship.

Aspects of inequity in health

Strata

- Household income, expenditure or wealth, sex, and place of residence (rural or urban)

Outcomes

- Risk, disease, death and social consequences of illness

Dimensions

- Access, quality and cost of treatment

Measures of health inequality

Simple measures

- Pairwise comparisons of health between two subgroups

Complex measures

- Use of data from all subgroups to assess inequality

Two types of inequality

- Absolute
- Relative

Simple measures of inequality

Absolute inequality

- Reflects the magnitude of difference in health between two subgroups
- Mean value of a health indicator in one subgroup subtracted from mean value of another subgroup

Relative inequality

- Reflects proportional differences in health among two subgroups
- Mean value of a health indicator in one subgroup divided by mean value of another subgroup

Complex measures of inequality

Absolute inequality

- Slope index of inequality

Relative inequality

- Ratio index of inequality
- Concentration index of inequality

Weighted for population size

Concentration index

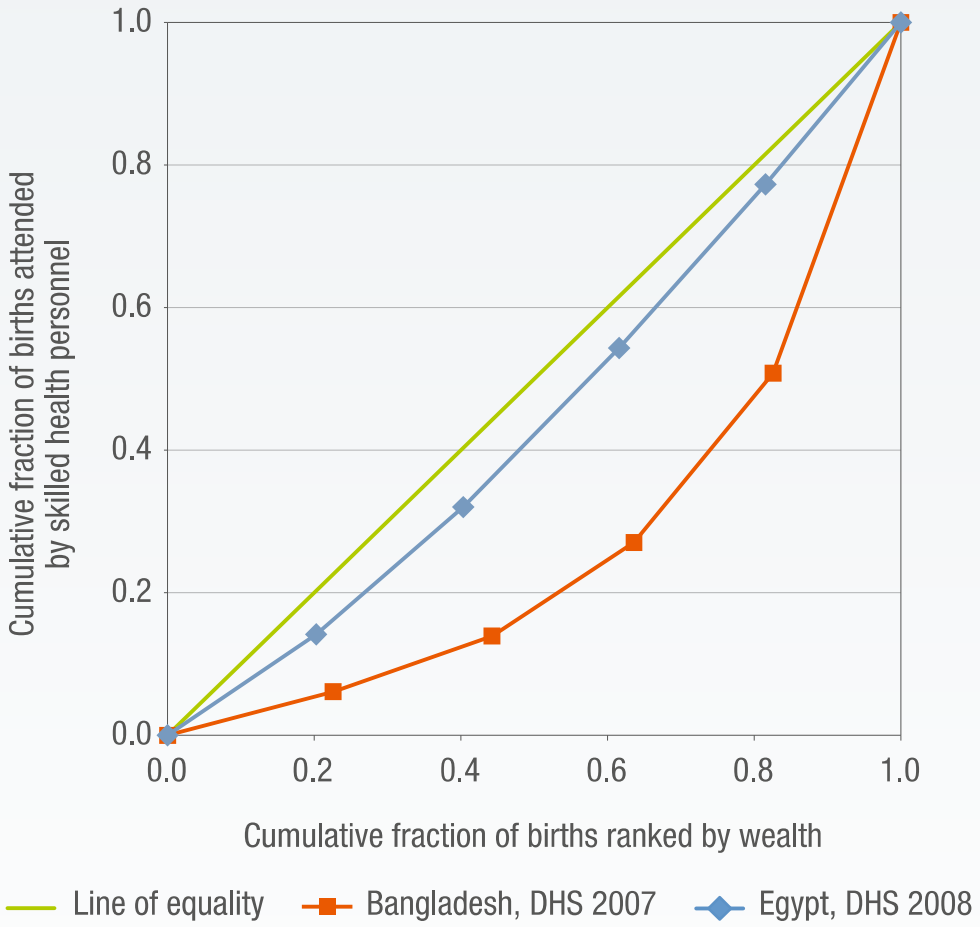
Relative measure of inequality

Extent to which a health indicator is concentrated among the disadvantaged or the advantaged

Range: -1 to +1 (-100% to +100%)

- Negative value: health indicator is concentrated among the disadvantaged
- Positive value: health indicator is concentrated among the advantaged
- Zero (0): no inequality

Example: concentration index



Relative wealth-based inequality in births attended by skilled health personnel in Bangladesh and Egypt, represented using concentration curves, DHS 2007 and 2008

Source: The World Health Organization (2013) Handbook on health inequality monitoring: with a special focus on low- and middle-income countries.

Measuring health inequalities

Equity stratifier	Simple measures of inequality		Complex measures of inequality	
	Absolute	Relative	Absolute	Relative
Wealth	Difference	Ratio	Slope index of inequality	Concentration index
Education	Difference	Ratio	Slope index of inequality	Concentration index
Area	Difference	Ratio		
Sex	Difference	Ratio		
Region	Difference	Ratio	Mean difference from overall mean	Theil index

Inequality in coverage, Bangladesh 2011

Indicators	Q1 (Poorest) (%)	Q5 (Richest) (%)	Difference (Q5 – Q1) (percentage points)	Ratio (Q5 / Q1)
Care seeking for pneumonia	56.6	78.8	22.2	1.39
At least four ANC with SBA	13.2	79.2	66.0	6.00
Pneumonia treatment	41.8	41.7	-0.1	1.00
BCG immunisation	70.6	97.3	26.7	1.38
Institutional delivery	31.1	84.7	53.6	2.72

Source: Bangladesh DHS 2011. The Demographic and Health Surveys (DHS) Program.

Summary

Aspects of Inequity

- Strata: groups over which inequity occurs
- Outcomes: disease or death states which are distributed unequally
- Dimensions: inequity in access, treatment quality or cost

Measures of health inequality

- Simple
- Complex
- Example: Concentration index

6. Aging

Basic Demographic Measures

Birth rate:

- Number of births per 1000 population

Fertility rate:

- Number of births a woman would have over her reproductive lifetime at current birth rates

Death rate:

- Number of deaths per 1000 population

Life expectancy at age X:

- Average length of time a person lives from age X at current death rates

Example: Japan, 2010

Population: 128,000,000

- <15 years: 16,800,000
- >65 years: 29,200,000 (23%)

Birth rate: 8.5 per 1000

Fertility rate: 1.39 per woman

Life expectancy: 83 years

The fastest-aging country in the world

Population Distribution

Shows the number of people in each age category
Pattern identifies where population is largest and smallest
Example: Japan, 2010

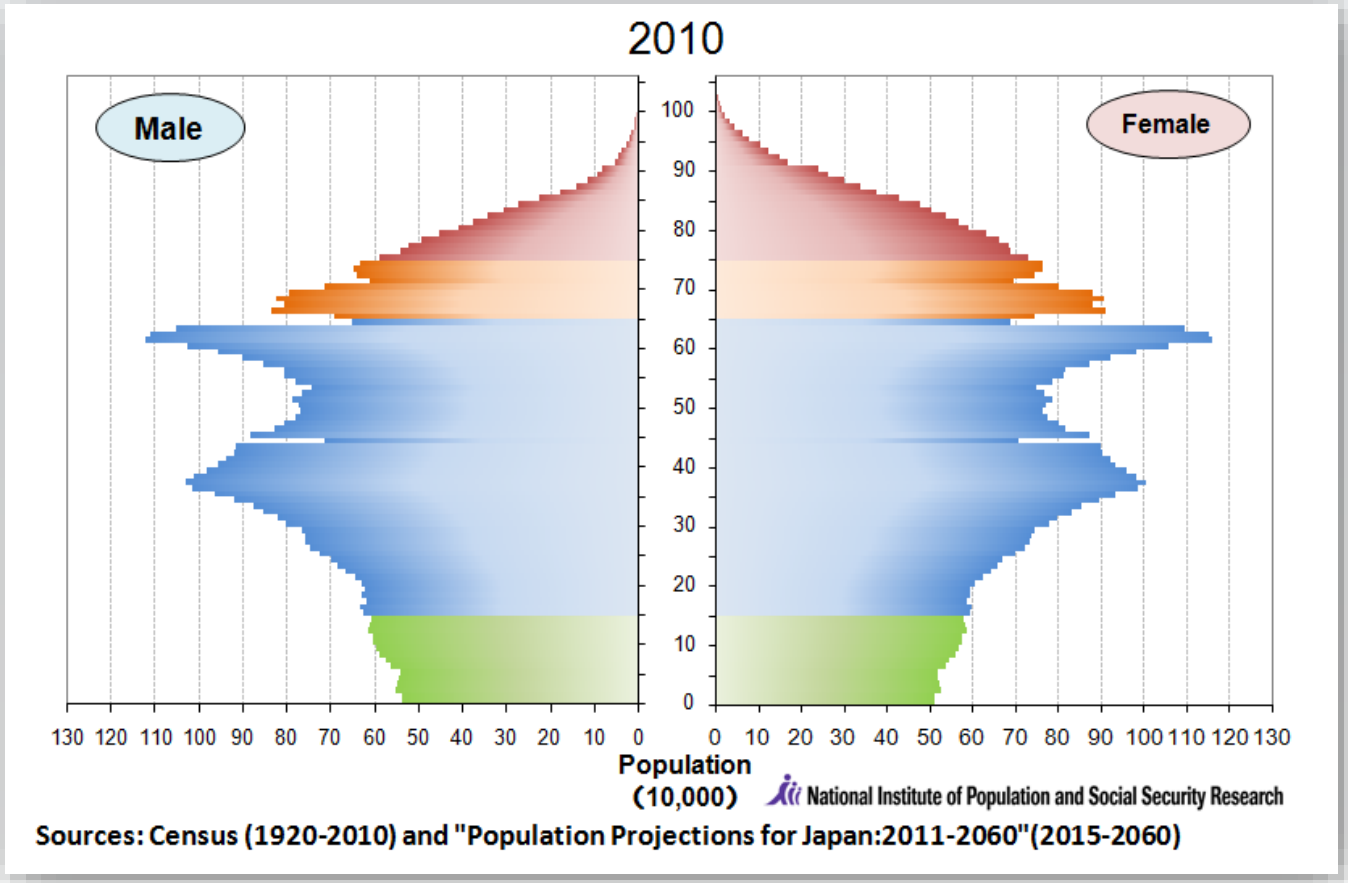


Image Source: IPSS website (http://www.ipss.go.jp/site-ad/TopPageData/2010e.png)

Measuring Aging

Specific population groups

- Working Population, Pw: aged 15-64 / 15-retirement age
- Elderly population, Pe: aged 65+/retirement +
- Child population, Pc: aged 0-14

Dependency ratios

- Measure relationship of demographic groups
- Dependency ratio (DR): (Pe+Pc)/Pw
- Old age dependency ratio (OADR): Pe/Pw

Give basic information on demographic trends

Measuring Aging

Dependency ratios tell us:

- Relative burden of care falling on working people
- Number of dependents each worker must support

Dependency ratios do NOT tell us:

- What proportion of Pw actually works
- What the effect of *non-demographic* policies will be

Common but imperfect measure of aging

How Does Aging Happen?

Aging is an inevitable consequence of development

- Improved healthcare -> lower mortality, longer lives
- Improved education -> lower birth rates, less working people

The Demographic transition:

- Growing proportion of elderly people
- Growing burden of non-communicable disease
- BUT infectious disease still prevalent
- Simultaneous burden of elderly care, NCDs, infectious disease

Aging and UHC

Aging is the return on investment in health

Aging threatens sustainability of UHC:

- Growing elderly population
- Growing burden of complex non-communicable disease
- Shrinking workforce
- Growing costs on a shrinking tax base

Health financing systems need to account for aging

Summary

- Aging can be measured using dependency ratios
- Aging is an inevitable consequence of improvements in health
- The demographic transition causes a shift from infectious to non-communicable disease
- Aging threatens sustainability of UHC systems
- UHC systems need to adapt to aging

7. Case study 1: Japan

Japan: Health Background

Japan has passed through the epidemiological transition

Health in Japan is characterized by:

- High life expectancy
- Low infant and maternal mortality
- Low burden of infectious disease
- High burden of non-communicable disease
- Predominance of behavioral risk factors

Japan's UHC system

Introduced in 1961

Social insurance-based system

- Membership is mandatory
- Payments as percentage of salary
- Reimbursement of 70% of medical costs

Highly regulated

- Price schedule fixed by government
- Strict regulatory framework

Japanese UHC: Achievements

Globally top-ranked in life expectancy

- 87 years for women, 81 years for men

Low-cost, equitable system

- Expenditure as % of GDP low compared to OECD
- Low levels of inequity in access
- Near universal coverage (>99%)

Rapid gains in major health issues

- 1960s: Maternal/child health
- 1980s: Stroke risks

Japan: The World's Fastest Aging Country

High proportion of elderly people

- 25% aged 65 years+
- 12% aged 75 years+

Rapid increase in elderly

- 40% of population aged over 65 by 2060
- Shrinking population and growing dependency

Difficulty sustaining UHC

- GDP spent on health: from 9.5% to 13% by 2030

Japanese UHC: Challenges

Ageing

- Shrinking workforce and tax base
- Increased demand for care of elderly
- Increased healthcare needs

Non-communicable diseases

- Increased burden of care
- Reduced productivity
- Complex, interacting medical needs

Both problems threaten sustainability

NCD Policy

Health promotion plans (1960s ->)

- Public health nurses
- Mass screening and health check ups
- Salt reduction strategies

Health Japan 21 (2001 ->)

- Introduction of quantitative targets
- Major reorientation to NCDs
- Greater focus on healthy life expectancy and living with illness

Future Policy Changes

Japan Vision 2035

- Fairness and equity in health care
- Support individual autonomy at all ages
- Shared global vision for NCD and aging policy

Integrated community care

- Supports community autonomy
- Coordinate care across doctors and facilities

Changes to insurance system

- Ensure stability and sustainability as population ages

Summary

- Japan saw huge health improvements in the 20th century
- Aging is the inevitable consequence of these improvements
- Japan is the fastest aging country
- Aging creates challenges for health financing
- Japan's UHC system requires reform to ensure sustainability

8. Case study 2: Bangladesh

Health Background

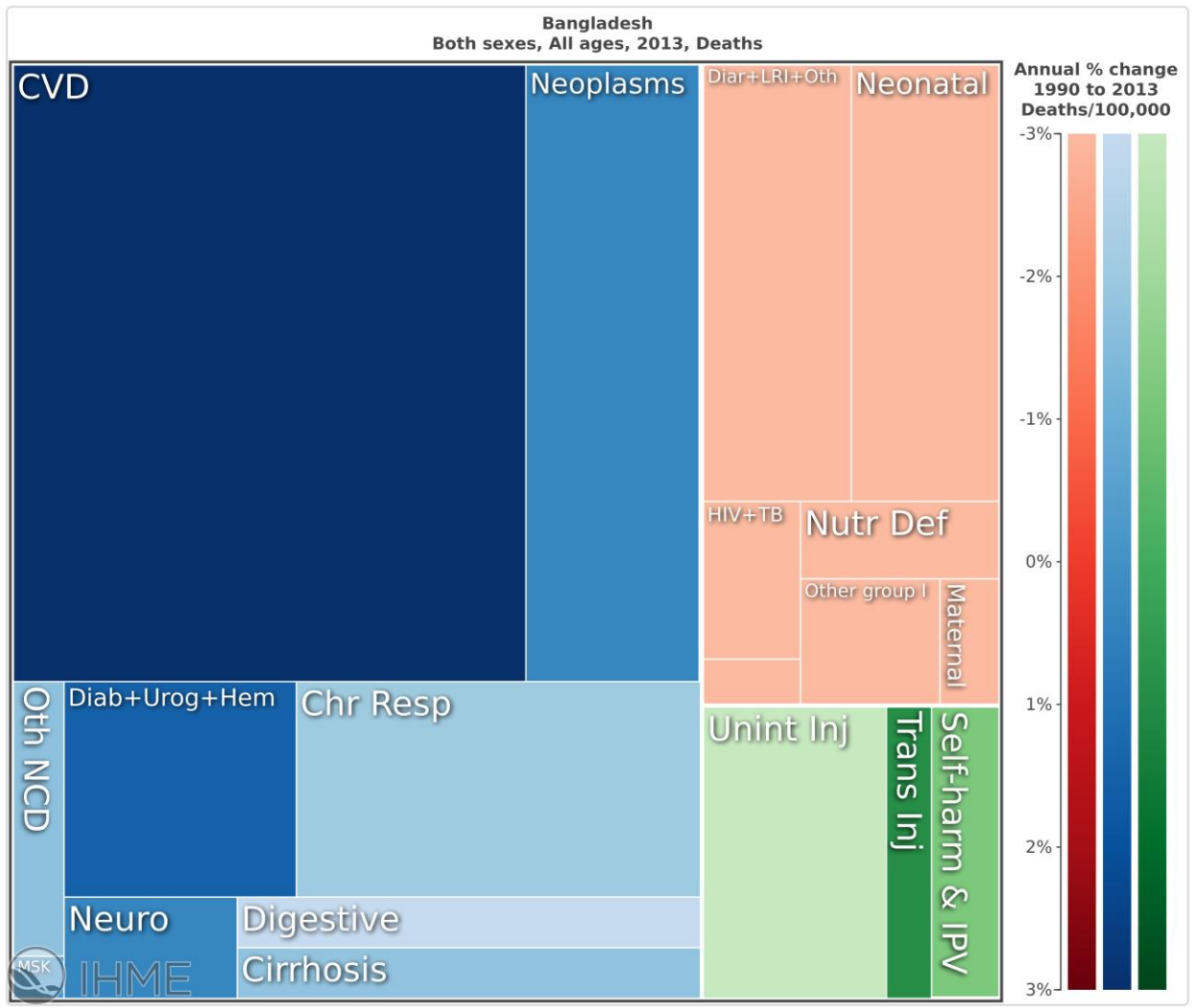
Bangladesh is a developing nation

- GDP: 747 US$ per capita (2014)
- Population: 159 million (2014)

Entering the epidemiological transition

- Maternal and child health a major issue
- Large burden of preventable infectious disease death
- Growing burden of non-communicable disease

Health Background



Source: Institute for Health Metrics and Evaluation (IHME). **GBD Compare**. University of Washington, 2015. Available from http://vizhub.healthdata.org/gbd-compare. CC BY-NC-ND 4.0.

Health Financing System

Bangladesh has no universal health coverage

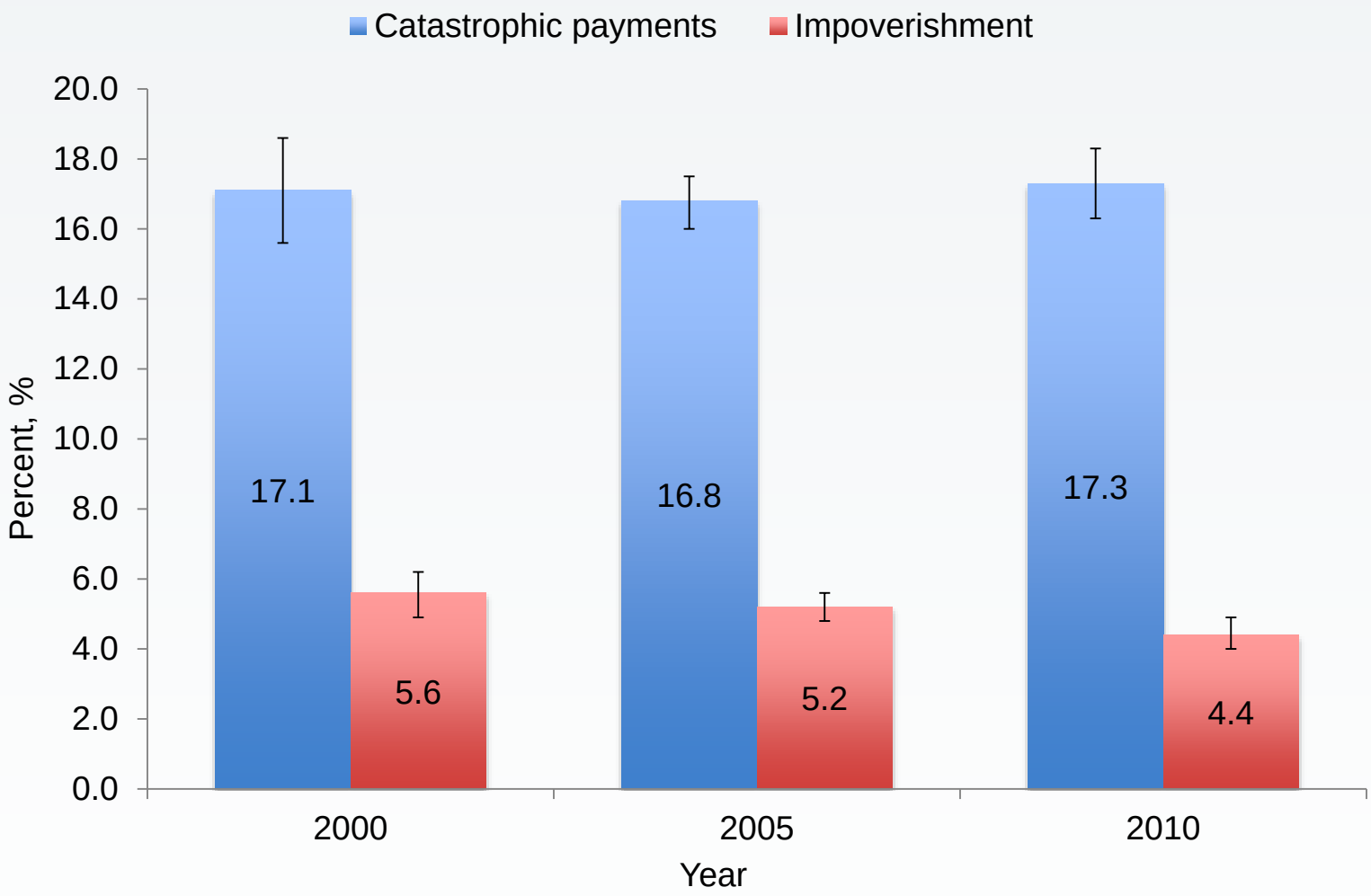
Major sources of financing

- Out of pocket payments: 64%
- Public funds: 26%
- International development partners: 10%

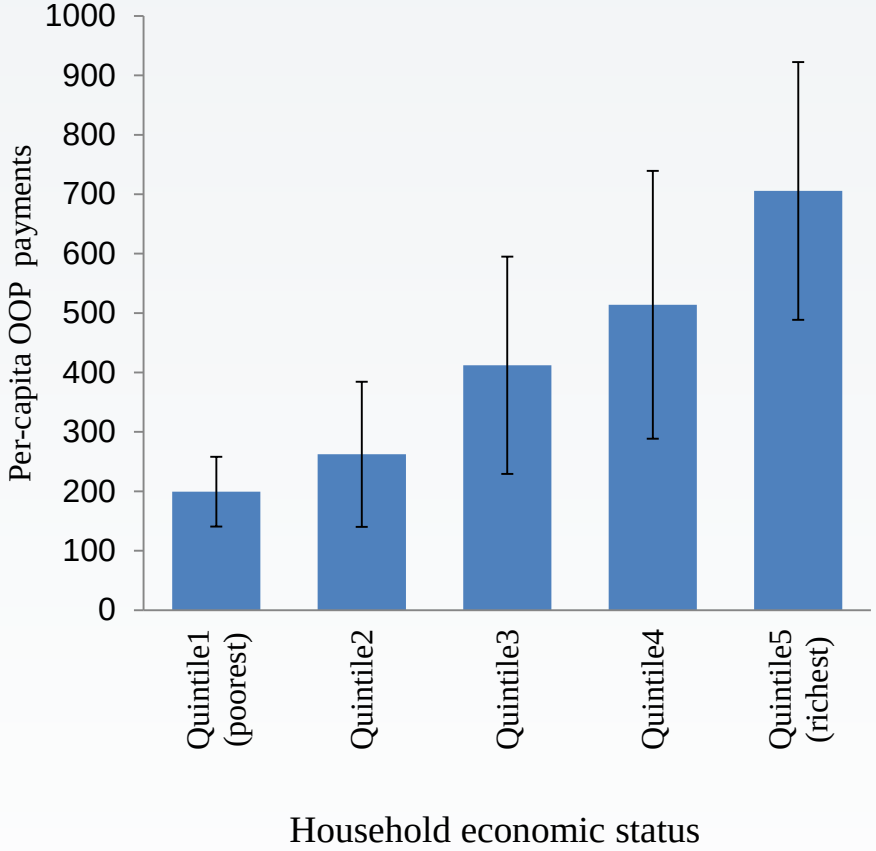
Catastrophic health payments are common

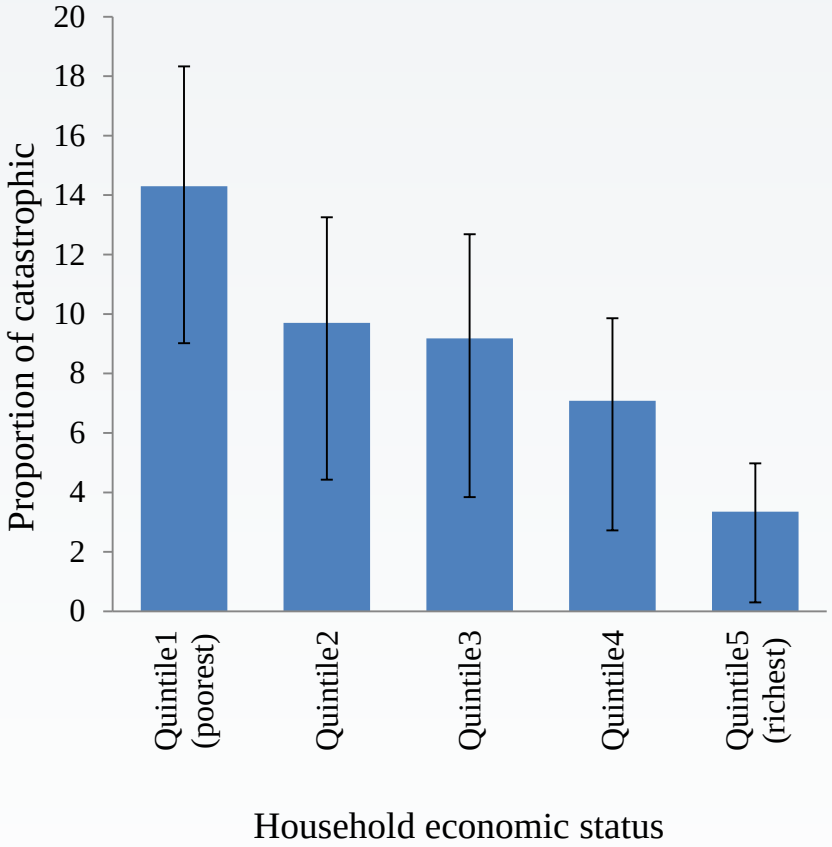
- 17% of the population
- Distress financing (borrowing or selling) is common (7%)

Financial risk associated with OOP payment



Health Financing Equity in Bangladesh





Progress towards UHC

Improved most maternal and child health related indicators

- Coverage rate >80%
- Improved water sources, no use of solid fuel, family planning need satisfied, vaccinations, and diarrhea treatment

Slightly declining impoverishment related to OOP payment

- Dropped to 4% in 2010 from 6% 2000
- But little change in OOP payments

Challenges to UHC Implementation

Poor health financing system

- Lack of formal health insurance
- High dependency on OOP payment
- Health small % of GDP

Low service coverage

- Skilled birth attendants and institutional delivery
- Treatment and control of NCD-related indicators

High inequality

- Service coverage and financial risk protection

A Typical Developing Nation Path

Typical of the challenges in developing nations

- Limited fiscal space
- Workforce and physical resource challenges
- Double burden of disease
- Weak regulatory frameworks

Solutions can be adapted from elsewhere

- UHC is a major goal of the UN and WHO
- Progress requires lessons from successful countries like Japan

Summary

Bangladesh is entering the demographic transition

- NCD disease burden increasing
- Infectious disease declining

Health financing system needs strengthening

- High incidence of catastrophic payments
- Inequality in health service coverage
- Inequality in financial risk protection

9. New policy directions

Policy Responses to Aging

Natalist policy

- Encourages childbirth

Immigration policy

- Draw young people from other communities

Productivity policy

- Change workforce structure

Natalist Policy

Suite of policies to increase the birthrate

- Enhance work/life balance (parental leave)
- Financial incentives (tax relief, bonuses)

Advantages:

- Domestic control over population
- Most people want children

Disadvantages:

- Take 15 years+ to take effect
- Require increased births -> against trend
- Increased youth dependency as <15 year old population grows

Immigration Policy

Policies to encourage net migration:

- Open borders (e.g. EU)
- Large migration schemes (Australia, Canada)

Advantages:

- Immediate effect
- Control over allocation of labor

Disadvantages:

- Can be politically unpopular (e.g. UK)
- Limited by language/culture (e.g. Japan, Korea)
- Limited by available pool of young migrants

Productivity Policy

Policies to adapt the workforce to aging

- Increase participation (e.g. women, elderly people)
- Increase workplace productivity
- Reorient workforce

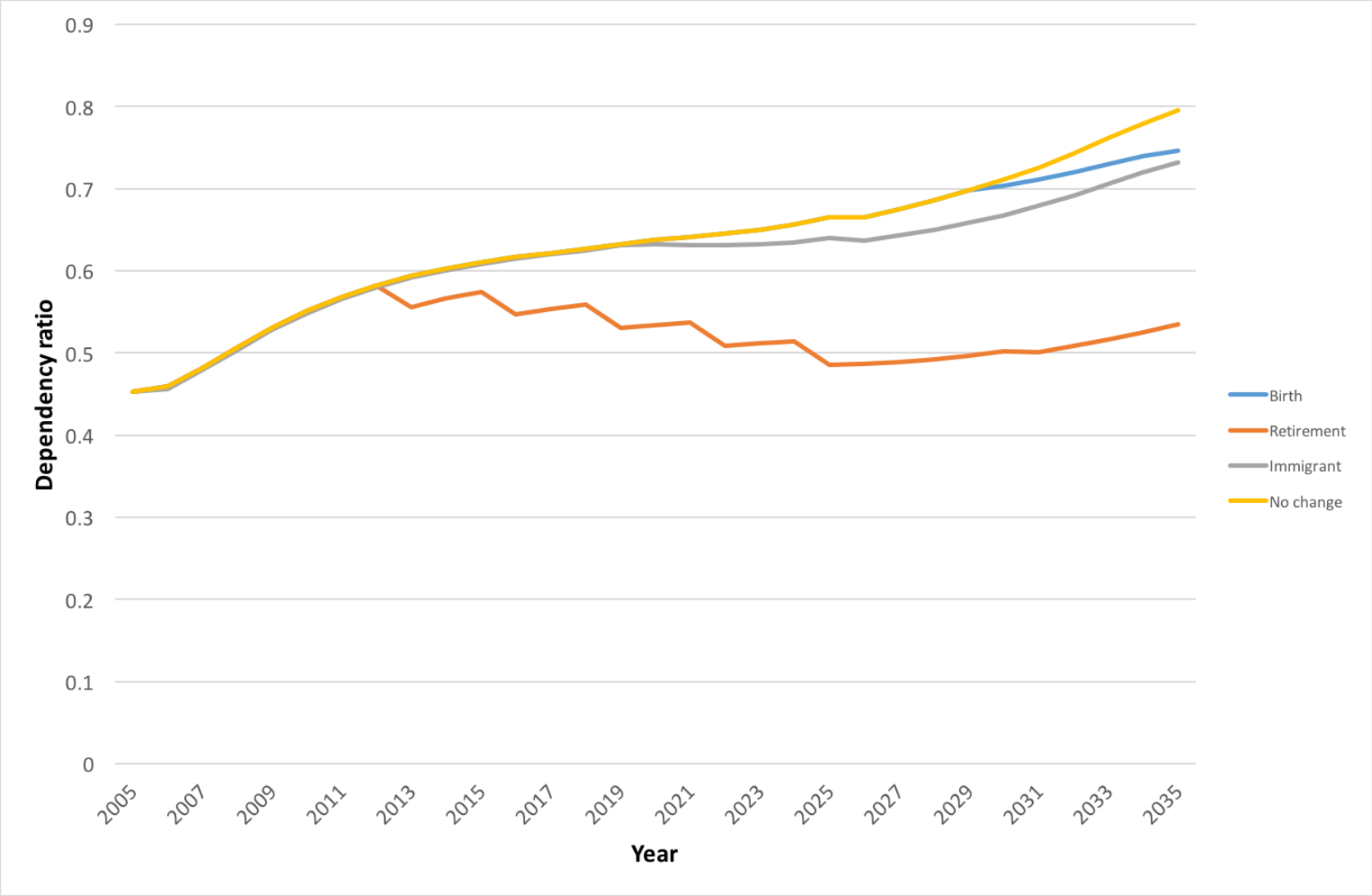
Advantages:

- Immediate effect
- Boost economic growth

Disadvantages:

- Require corporate/industrial cooperation
- Effectiveness limited by social/economic factors

Example: Three Policies in Japan



Aging as a Health Workforce Issue

Increased burden of elderly care:

- Reduced birth rate -> Single child families -> loss of private care work force
- Increased elderly population -> Increased care needs

Increased NCD burden:

- Greater need for complex, chronic care
- Increased ancillary staff needs (e.g. rehabilitation)

Direct workforce impact (less workers, inflation)

Financial pressures (less payers, more payees)

Healthy Aging and the Health Workforce

Aging societies focus on healthy aging

- Aim to maintain independent living
- Benefits individual health and reduces workforce burden

Policy changes for healthy aging:

- Health promotion to protect mobility and lifestyle
- Health care to manage NCDs
- Urban planning to facilitate autonomy

Requires cross-sectoral integrated care policies

Integrated Care Policy

Traditional focus of health system

- Curative care
- Disease-specific focus

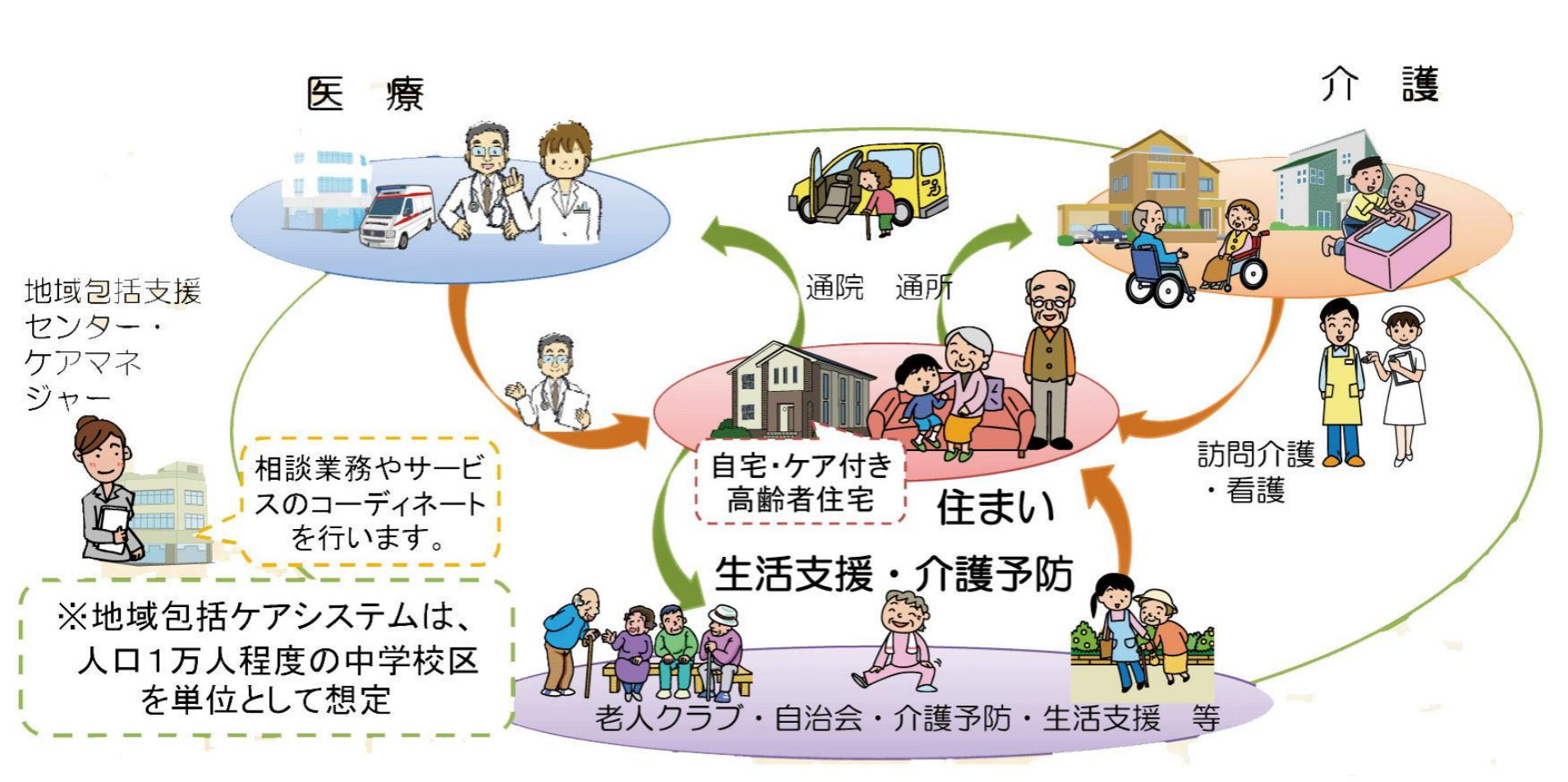
After the epidemiological transition

- Complex, interacting NCDs
- Multiple care agents (family, cardiologist, community nurse, care facility)

Need to integrate care at the community level

Streamline health financing for new systems

Example: Japan's Integrated Care Policy



Source: Ministry of Health, Labour and Welfare, Japan

Example: New Financing for the ”Old Old”

Health financing arranged at prefecture level

- Small prefectures have large elderly population
- Cross-subsidized from younger prefectures
- Subsidies support care of those over 65

Growing population -> pressure on this system

Reformed in 2008

- Reduced subsidies to the “young old” (65-74)
- Increased subsidies to the “old old” (75+)

Ensure stability of social insurance-based system

Summary

Three policy responses to aging

- Natalist, immigration, productivity
- Balance of effectiveness, timeliness and acceptability

Aging is a workforce as well as a demographic issue

- Change in workforce composition
- Changes in tax base and scale of workforce

Respond with health system reforms

- Change to structure and orientation of system
- New financing methods